Дз

№1 $n=15$ $d=5$ $GF(2^4)$

$g(x)=(x-\alpha)(x-\alpha^2)(x-\alpha^3)(x-\alpha^4)=(x^2-\alpha^2x-\alpha x+\alpha^3)\cdot$

$\cdot(x^2-\alpha^4x-\alpha^3x+\alpha^7)=(x^2-\alpha^5x+\alpha^3)(x^2-\alpha^7x+\alpha^7)=$

$=x^4-\alpha^7x^3+\alpha^7x^2-\alpha^5x^3+\alpha^{12}x^2-\alpha^{12}x+\alpha^3x^2-\alpha^{10}x+\alpha^{10}=$

$=x^4+\alpha^{13}x^3+\alpha^6x^2+\alpha^3x+\alpha^{10}$

№2

$PC(75, 9)$ $GF(2^4)$ $d = 7$

$p(x) = 1 + x + x^4$

$v(x) = \alpha + \alpha^3 x + \alpha^{70} x^2 + \alpha^{72} x^3 + \alpha^{73} x^4 + \alpha^3 x^5 + \alpha^9 x^6 + \alpha^5 x^7 +$

$+ x^8 + \alpha^{70} x^9 + \alpha^8 x^{10} + \alpha x^{11} + \alpha x^{12} + \alpha^{73} x^{13} + \alpha^7 x^{14}$

$S_i = \sum_{j=0}^{14} v_j (\alpha^i)^j$

$S_1 = \alpha + \alpha^3 \alpha + \alpha^{70} \alpha^2 + \alpha^{72} \alpha^3 + \alpha^{73} \alpha^4 + \alpha^3 \alpha^5 + \alpha^9 \alpha^6 + \alpha^5 \alpha^7 +$

$+ \alpha^8 + \alpha^{70} \alpha^9 + \alpha^8 \alpha^{10} + \alpha \alpha^{11} + \alpha \alpha^{12} + \alpha^{73} \alpha^{13} + \alpha^7 \alpha^{14}$

$= \alpha^2 + \alpha^{72} + \alpha^{73} + \alpha^{77} = x^2 + x^3 + x^2 + x^3 = 0$

— // — $S_2 = \ldots = S_6 = 0$

$\Rightarrow e(x) = 0$